# Coding Discussion: Recursive Thinking

**Design and Analysis of Algorithms**

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# I Thought About Writing Some Code in Scheme…

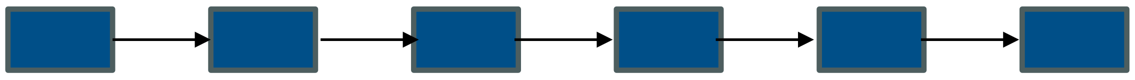
… but then I reconsidered.

$$N! = N \times \underbrace{\boxed{(N-1) \times (N-2) \times (N-3) \times \ldots \times 2 \times 1}}_{(N-1)!}$$

```
(define (fact N) (if (= N 0) 1 (* N (fact (- N 1)))))
```

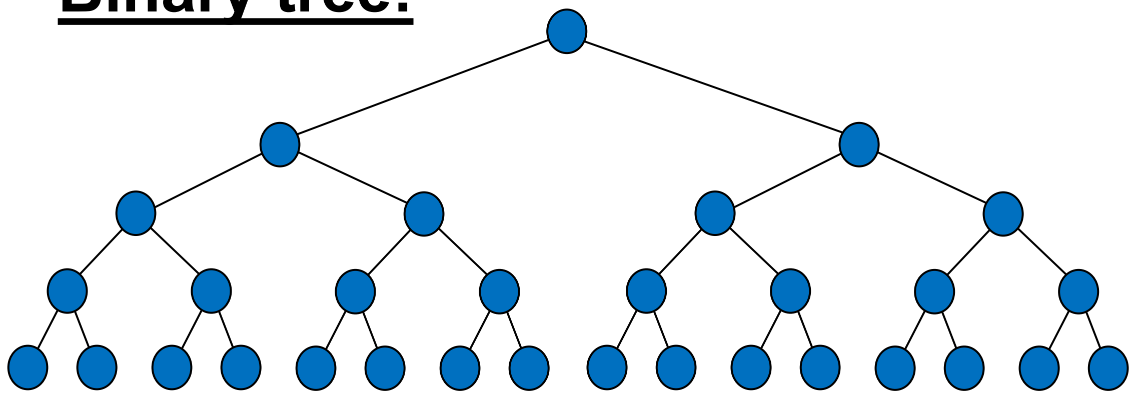
# Recursive Structure in Common Data Structures

**Linked list:**



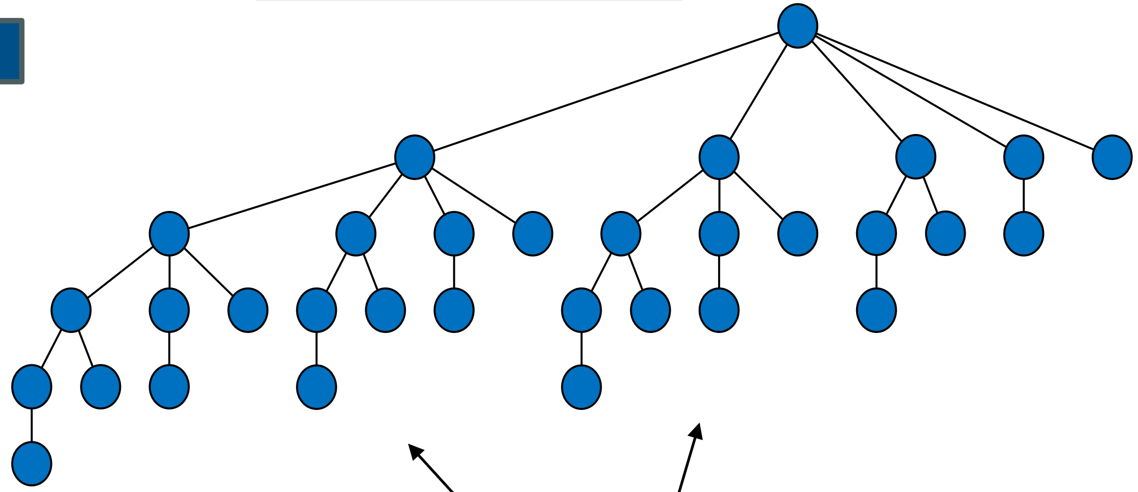
Also a linked list!

**Binary tree:**



Also binary trees!

**Binomial tree:**



Also binomial trees!

# Recursive Structure in Common Data Structures

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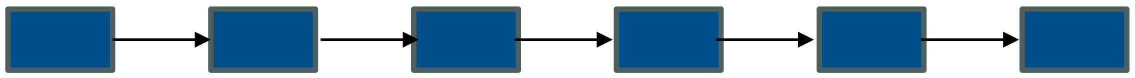
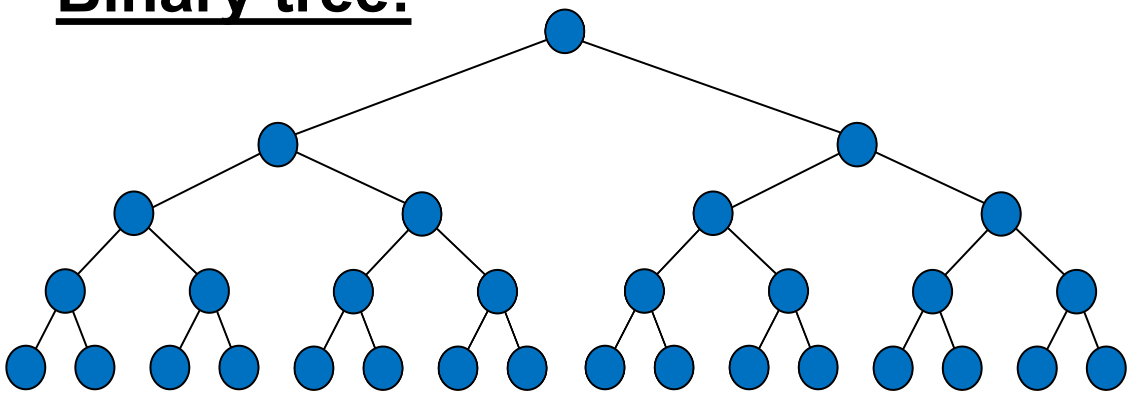


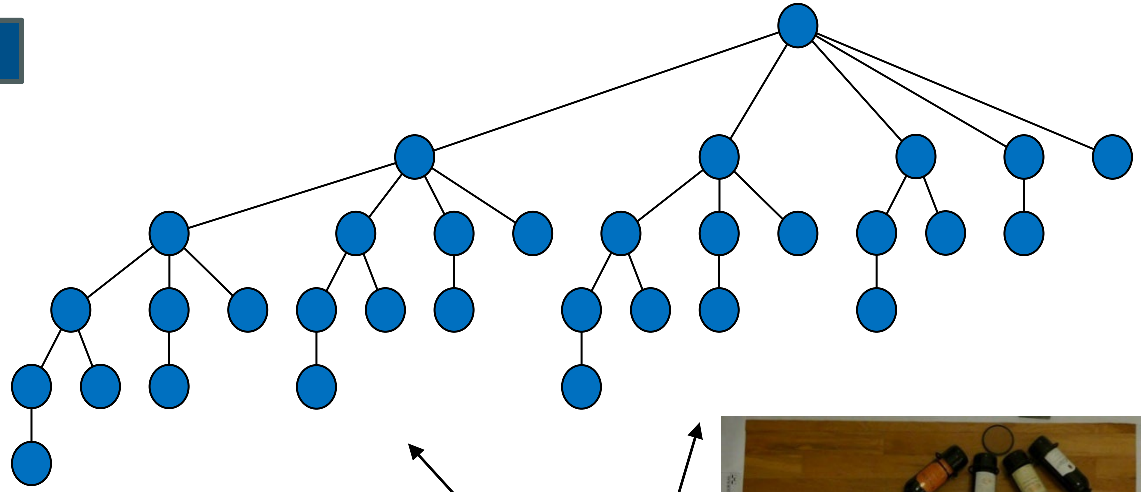
Also a linked list!

**Binary tree:**



Also binary trees!

**Binomial tree:**



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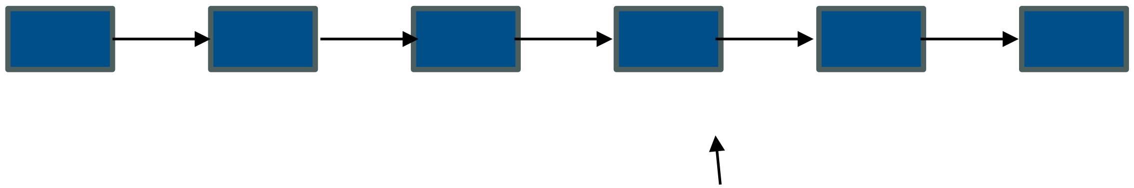
# Today’s Goal

- Practice thinking and coding recursively.
- No loops allowed!
- For simplicity, we’ll work with linked lists
- Main Tasks:
  - Reverse a linked list
  - Sort a linked list
    (insertion sort, merge sort)



Also a linked list!